

Introduction to Contract Theory and Auctions
Lecture 7
Screening: Applications

Prof. Alejandro Melo-Ponce



NAZARBAYEV
UNIVERSITY

ECON 403/503
Fall 2021

- Analyzed in two contexts:
 - **Procurement:** The firm supplies a good to the government.
 - Defense Contracts.
 - **Regulation:** The firm supplies a good to consumers on behalf of government.
 - Utilities (electricity, gas), telecommunications, railroads.
- There are many natural monopolies in utilities, telecommunication, infrastructure, and transportation industries.
 - Due to large fixed costs, average cost is decreasing even at large production levels.
 - It may be socially efficient to have a monopoly.
- Government needs to regulate to prevent them acting as a monopolist.
- But it doesn't know the relevant characteristics (e.g., technology) and/or does not observe actions (e.g., cost reducing effort) of the firm.

- Typically, government uses accounting (cost) data and reimburses a fraction of firm's monetary expenditures c .
- Its aim is to obtain the highest societal welfare in providing the good.
- We will model it as follows: government pays c and a net transfer t to the firm

$$t = a - bc$$

- a : a fixed fee.
- b : fraction of cost borne by the firm (measures the power of incentives).
- Possible contracts
 - **cost-plus (cost-of-service)**: $b = 0$
 - firm does not bear any cost.
 - low powered: firm does not have an incentive to lower cost.
 - **fixed-price (price cap)**: $b = 1$
 - government does not reimburse any cost; pays only a fixed fee.
 - high powered: firm is the residual claimant for its cost savings.
 - **incentive (cost-sharing)**: $0 < b < 1$

- Procurement 1960 US:

40.9%	cost-plus
13.6%	incentive
31.4%	fixed-price

- Regulation:

- historically: cost-of-service.
- more recently: price-cap and incentive regulation.

- Government: provide incentives to reduce cost and extract firm's rent.
- Fixed-price
 - Incentives: Good
 - Firm is the residual claimant.
 - Provides efficient effort.
 - Rent extraction: Bad
 - Rent is non-negative when cost is high.
 - Exogenous reduction in cost benefits the firm.
- Cost-plus
 - Incentives: Bad
 - Rent Extraction: Good
- Optimal contract
 - Perfect information about technology: fixed-price
 - Imperfect information: incentive (cost-sharing) contract

- Early models
 - Baron, D. and R. Myerson (1982), “Regulating a Monopolist with Unknown Costs”, *Econometrica*, 50, 911-30.
 - Laffont, J.-J. and J. Tirole (1986), “Using Cost Observation to Regulate Firms”, *Journal of Political Economy*, 94, 614-41.
- Main reference: Laffont, J.J. and J. Tirole (1993), *A Theory of Incentives in Procurement and Regulation*, Cambridge: MIT Press.

A simple model of regulation

- Our model: a simple version of Laffont and Tirole (1986).
- The firm produces a single indivisible output.
- Accounting cost is $c = \theta - e$.
 - $\theta \in \{\theta_L, \theta_H\}$, $\Pr(\theta_L) = p$.
 - e is effort which costs the firm $\psi(e) = e^2/2$.
 - We assume $0 < \Delta\theta = \theta_H - \theta_L \leq 1 - p$
- Regulator wants to maximize $S - P$
 - S is (constant) consumer surplus from consuming the good.
 - $P = c + t$ is total payment, t is net transfer to the firm.
- Therefore, regulator wants to minimize P .
- Payoff of the firm $P - c - e^2/2 = t - e^2/2$.
- Reservation payoff of the firm is zero.
- c is observable but not θ or e .

- Regulator knows θ (therefore, can deduce $e = \theta - c$).
- Its problem is:

$$\min_{(e,t)} \{\theta - e + t\} \quad \text{s.t.} \quad t - e^2/2 \geq 0, \quad e \geq 0.$$

- The constraint must bind: $t = e^2/2$

$$\min_{e \geq 0} \{\theta - e + e^2/2\}$$

- The first order condition is

$$-1 + e^* \leq 0$$

- If the solution is interior ($e^* > 0$), the first order condition holds with equality

$$-1 + e^* = 0 \Rightarrow e^* = 1$$

- Comparing with the boundary ($e = 0$) shows that the solution must be interior.

- Effort is efficient

$$\text{marginal cost} = \psi'(e^*) = e^* = 1 = \text{marginal benefit}$$

and

$$t^* = 0.5 \quad \text{firm receives no rent}$$

- Regulator pays the firm a fixed total payment $P = \theta - 0.5$
- Induces the firm to minimize $c + e^2/2$.
- This is a price-cap (fixed-fee) regulation

$$t = a - (c - c^*)$$

where

$$\begin{aligned} a &= \psi(e^*) = 0.5 \\ c^* &= \theta - e^* = \theta - 1 \end{aligned}$$

- What happens if regulator offers first best?
 - $c_L = \theta_L - 1$, $c_H = \theta_H - 1$ and $t_L = t_H = 0.5$
- For the efficient type (θ_L)
 - Effort required to obtain cost c_L is 1.
 - Effort required to obtain cost c_H (e') is $1 - \Delta\theta < 1$:

$$c_H = \theta_H - 1 = \theta_L - e'$$

- Since transfers are the same, she prefers to mimic the inefficient type.
- Usual recipe for optimal contract:
 - Efficient type (θ_L):
 - provides first best effort
 - obtains positive rent
 - Inefficient type (θ_H)
 - provides less than first best effort (to reduce efficient type's rent)
 - obtains no rent

Unobservable θ and e

- Regulator chooses a contract $\{(t_L, c_L), (t_H, c_H)\}$ to minimize $p(t_L + c_L) + (1 - p)(t_H + c_H)$ subject to IR and IC constraints.
- Since $c = \theta - e$, the problem can be written as

$$\begin{aligned} \min_{(t_i, e_i)_{i=H,L}} \quad & p(t_L + \theta_L - e_L) + (1 - p)(t_H + \theta_H - e_H) \\ \text{subject to} \quad & t_L - e_L^2/2 \geq 0 \\ & t_H - e_H^2/2 \geq 0 \\ & t_L - e_L^2/2 \geq t_H - (e_H - \Delta\theta)^2/2 \\ & t_H - e_H^2/2 \geq t_L - (e_L + \Delta\theta)^2/2 \end{aligned}$$

- Note that for type θ_L , effort required to achieve cost $c_H = \theta_H - e_H$ is equal to $e_H - (\theta_H - \theta_L)$. Similarly, for high type's effort to achieve low cost.
- We implicitly assume $e_H - \Delta\theta \geq 0$, which we will need to verify once we solve the problem.

Solving the problem

- ① IR_L does not bind:

$$t_L - e_L^2/2 \geq t_H - (e_H - \Delta\theta)^2/2 > t_H - e_H^2/2 \geq 0$$

- ② IC_L binds, otherwise

$$t_L - e_L^2/2 > t_H - (e_H - \Delta\theta)^2/2 > t_H - e_H^2/2 \geq 0$$

and the regulator can decrease t_L

- ③ IR_H binds, otherwise, by step 1, can decrease t_L and t_H by the same small amount.
- ④ $e_H \leq e_L + \Delta\theta$ or $c_H \geq c_L$: IC_L , IC_H imply

$$e_H^2/2 - (e_L + \Delta\theta)^2/2 \leq t_H - t_L \leq (e_H - \Delta\theta)^2/2 - e_L^2/2$$

Result follows from simple algebra.

- ⑤ IC_H can be ignored: follows from steps 2 and 4.

- Therefore, the problem is equivalent to

$$\begin{aligned} \min_{(t_i, e_i)_{i=H,L}} \quad & p(t_L + \theta_L - e_L) + (1 - p)(t_H + \theta_H - e_H) \\ \text{subject to} \quad & t_H - e_H^2/2 = 0 \\ & t_L - e_L^2/2 = t_H - (e_H - \Delta\theta)^2/2 \end{aligned}$$

- Note that

$$t_L - e_L^2/2 = \Delta\theta(e_H - \Delta\theta/2)$$

- $e_H \geq \Delta\theta$ implies that type θ_L obtains a rent.

Solving the problem

Substituting into the objective the constraints, the problem is equivalent to

$$\min_{e_L, e_H \geq 0} \{p(e_H^2/2 + e_L^2/2 - (e_H - \Delta\theta)^2/2 + \theta_L - e_L) + (1-p)(e_H^2/2 + \theta_H - e_H)\}$$

or

$$\max_{e_L, e_H \geq 0} -\{p(e_H^2/2 + e_L^2/2 - (e_H - \Delta\theta)^2/2 + \theta_L - e_L) + (1-p)(e_H^2/2 + \theta_H - e_H)\}$$

Note that the term $-(p\theta_L + (1-p)\theta_H)$ appears in the expression above, and it is independent of either e_L or e_H . Thus, the problem is equivalent to:

$$\max_{e_L, e_H \geq 0} -\{p(e_H^2/2 + e_L^2/2 - (e_H - \Delta\theta)^2/2 - e_L) + (1-p)(e_H^2/2 - e_H)\}$$

- The objective function is strictly concave and the constraint functions e_L and e_H are concave.
- Therefore, if a solution exists, the Kuhn-Tucker conditions will uniquely identify it.
- The Lagrangean is given by

$$\mathcal{L} = -\{p(e_H^2/2 + e_L^2/2 - (e_H - \Delta\theta)^2/2 - e_L) + (1-p)(e_H^2/2 - e_H)\} + \lambda_L e_L + \lambda_H e_H$$

Solving the problem

- Critical points are given by

$$\begin{aligned} -p(e_L - 1) + \lambda_L &= 0 \\ -(p\Delta\theta + (1 - p)(e_H - 1)) + \lambda_H &= 0 \\ \lambda_L, \lambda_H, e_L, e_H &\geq 0, \quad \lambda_L e_L = \lambda_H e_H = 0 \end{aligned}$$

- First equation and $\lambda_L \geq 0 \Rightarrow e_L > 0 \Rightarrow \lambda_L = 0$, which implies

$$e_L = 1$$

- Similarly the assumption $\Delta\theta \leq 1 - p$ implies that

$$e_H = 1 - \frac{p}{1 - p} \Delta\theta$$

- and $e_H \geq \Delta\theta$, as required.

- Familiar “no distortion at the top” and “under-provision for inefficient types” result.
- Optimal regulation: offer a menu so that
 - Efficient firm chooses a price-cap regulation.
 - A constant payment $P_L = t_L + \theta_L - e_L$
 - $t_L = a_L - b_L c$, $a_L = t_L + \theta_L - e_L$ and $b_L = e_L = \psi'(e_L) = 1$
 - Inefficient firm chooses a cost-sharing arrangement
 - $t_H = a_H - b_H c$, $a_H = t_H + e_H(\theta_H - e_H)$ and $b_H = e_H = \psi'(e_H) < 1$

- Why are certain worthy projects not funded?
- If there is excess demand of funds, the interest rate should increase to equate demand with supply.
- The reason might be **adverse selection**.
 - As the interest rate increases only riskier projects apply.
- Relevant reference: Stiglitz, J. and A. Weiss (1981)
- In general, financial markets are prone to adverse selection.
- Lender knows less about the risk-return of a project than the borrower.
- Lender is like a *used-car buyer*.
- Inefficiencies can arise in the form that *good quality projects* remain *unsold*, i.e. they are denied credit. → **credit rationing**.

- Consider a population of risk-neutral borrowers.
- Each borrower owns a project which requires an initial investment of $I = 1$.
- Each project yields a random return $X \in \{R, 0\}$.
- p denotes the probability that $X = R$.
- Borrowers have no wealth, so they must obtain funds from an outside source.
- A borrower can be of two types, $i = s, r$, s : safe, r : risky.
- A borrower of type i has a project with return characteristics (p_i, R_i) .

- Assumptions

A1 $p_i R_i = m$ with $m > 1$.

A2 $p_s > p_r$ and $R_s < R_r$.

- Both Borrowers have same expected return, but different risk profile.
- A bank can offer to finance the initial investment in exchange for future repayment.
- Suppose there is only one bank, this bank has an **excess demand for funds**: amount available for funding is $\alpha < 1$. (Why we need this?)
- Unit mass of borrowers: proportion of safe borrowers is β and of risky borrowers is $1 - \beta$.
- Assume also that $\alpha > \max\{\beta, 1 - \beta\}$ so that available funds are sufficient to avoid crowding out either type of borrower.
- The first best contract (complete, symmetric information) is quite simple:
 - Bank is indifferent to whom to lend and lends the full amount α of funds
 - The contract specifies the repayment: D_i with $D_i = R_i$.
 - Whomever it lends to, repayment happens with probability p_i .

- Suppose there is adverse selection.
- Initial literature focused on quite simple contracts: **single fixed-repayment** schemes.
- A single-fixed repayment scheme specifies an amount to be repaid D in exchange for the initial outlay $I = 1$.
- Assume that type- i borrowers apply for funds if and only if $D \leq R_i$.
- If $D > R_s$, only risky borrowers will consider applying.
 - In this case, the bank sets $D = R_r$ which gives a profit of $(1 - \beta)(p_r R_r - 1) = (1 - \beta)(m - 1)$.
- If $D \leq R_s$, both types of borrowers apply. Assuming each applicant has an equal chance of being financed (by a law of large numbers)
 - The bank finds it optimal to set $D = R_s$
 - Obtains a profit of $\alpha[\beta(m - 1) + (1 - \beta)(p_r R_s - 1)]$.

Credit Rationing arises under single fixed Repayment

- Note that $D = R_s$ is better than $D = R_r$ if and only if

$$\frac{\alpha\beta}{1-\beta} + \frac{\alpha(p_r R_s - 1)}{m-1} > 1$$

- If this is the case, then **credit rationing arises**: some risky borrowers cannot get credit.
- The risky borrowers are willing to accept a higher D , but the bank finds optimal not to raise D because it would lose the safe borrowers.
- Remark: This is an example of a **single-price** contract.
- Can the bank do better by offering more sophisticated contracts?

Type dependent fixed repayment contracts—Second Best

- A contract is defined by (x_i, D_i) where x_i is the probability of obtaining financing and D_i is the amount to be repaid.
- The bank solves the following problem (P)

$$\max_{(x_i, D_i)_{i=s,r}} \{ \beta x_s (p_s D_s - 1) + (1 - \beta) x_r (p_r D_r - 1) \} \quad (1)$$

subject to

$$0 \leq x_i \leq 1, \quad \forall i = s, r \quad (\text{Feasibility 1})$$

$$\beta x_s + (1 - \beta) x_r \leq \alpha \quad (\text{Feasibility 2})$$

$$x_i p_i (R_i - D_i) \geq 0 \quad \forall i = s, r \quad (\text{PC})$$

$$x_i p_i (R_i - D_i) \geq x_j p_j (R_i - D_j) \quad \forall i, j = s, r \quad (\text{IC})$$

- If the bank offers the first best menu $\{(x_s, D_s), (x_r, D_r)\} = \{(\alpha, R_s), (\alpha, R_r)\} \rightarrow$ Risky type mimics the safe type.

Solving the problem

- The previous consideration suggests the reduced problem (RP):

$$\max_{(x_i, D_i)_{i=s,r}} \{ \beta x_s (p_s D_s - 1) + (1 - \beta) x_r (p_r D_r - 1) \} \quad (2)$$

subject to

$$x_s p_s (R_s - D_s) \geq 0 \quad (\text{PC safe})$$

$$x_r p_r (R_r - D_r) \geq x_s p_r (R_r - D_s) \quad (\text{IC risky})$$

$$x_r \geq x_s \quad (\text{Monotonicity})$$

$$0 \leq x_i \leq 1 \quad i = s, r \quad (\text{Feasibility 1})$$

$$\beta x_s + (1 - \beta) x_r \leq \alpha \quad (\text{Feasibility 2})$$

- The reduced problem should be familiar:
 - PC constraint for the risky type does not bind.
 - IC constraint for the safe type does not bind.
 - Monotonicity of the allocation.
- The solution of (P) is equivalent to the solution of (RP).

Proof that P is equivalent to RP

- ① Constraints of (P) imply those of (RP): From the two IC constraints:

$$x_r R_s - x_s R_s \leq x_r D_r - x_s D_s \leq x_r R_r - x_s R_r$$

which implies $x_r \geq x_s$.

- ② At the solution to (RP), IC constraint holds as equality: otherwise the bank can increase D_r

- Here we use $x_r > 0$. If $x_r = 0$ then by step 1, $x_s = 0$, which cannot be a solution to (RP).

- ③ Solution of (RP) satisfies the constraints of (P)

- (PC safe) and (IC risky) constraints for (P) imply PC for the risky type:

$$x_r p_r(R_r - D_r) \geq x_s p_r(R_r - D_s) \geq x_s p_r(R_s - D_s) \geq 0$$

- Step 2 and monotonicity imply IC for safe type.

$$\begin{aligned} p_s(x_s(R_s - D_s) - x_r(R_s - D_r)) &= p_s(x_s R_s - x_r R_s + x_r D_r - x_s D_s) \\ &= p_s(x_r - x_s)(R_r - R_s) \\ &\geq 0 \end{aligned}$$

- Suppose that $x_s^* > 0$. From (PC safe), which must hold with equality, we get $D_s^* = R_s$.
- Substitute binding (IC risky) and (PC safe) into the objective function:

$$\max_{x_s, x_r} x_s (\beta(p_s R_s - 1) - (1 - \beta)p_r(R_r - R_s)) + (1 - \beta)x_r(p_r R_r - 1)$$

- Since we assumed $p_i R_i = m > 1$ at an optimal solution

$$x_r^* = 1 > \alpha \quad \text{why?}$$

- $x_s^* > 0$ implies

$$\beta(p_s R_s - 1) - (1 - \beta)p_r(R_r - R_s) \geq 0 \quad \text{why?}$$

- Together with monotonicity and (Feasibility 2) we get

$$x_s^* = \frac{\alpha - (1 - \beta)}{\beta} < \alpha$$

- IC for risky type $(R_r - D_r^*) = x_s^*(R_r - D_s^*)$ and $D_s^* = R_s$ imply

$$D_r^* = x_s^* R_s + (1 - x_s^*) R_r > R_s = D_s^*$$

- If $\beta(p_s R_s - 1) - (1 - \beta)p_r(R_r - R_s) < 0$ then

$$x_s^* = 0$$

Solution: no credit rationing

- At this solution the bank trades off the rents extracted from risky borrowers with the ability to lend all its funds.
- When safe borrowers are numerous enough, bank wants to lend to them which leads to rent for risky borrowers.
- But rather than financing everybody by setting $D = R_s$ as in the single-price contract, the bank gives risky borrowers preferential access by setting $x_r^* > x_s^*$ in exchange for higher repayment.
- Credit rationing disappears: the only ones not fully funded are safe borrowers, but they are indifferent about being funded.
- Credit rationing reappears if we instead assume $p_r R_r < 1 < p_s R_s$.
 - Bank wants to turn away risky borrowers (i.e. set $x_r = 0$), but cannot do it without also denying the safe ones (since $x_r \geq x_s$).
 - Either everybody is funded (if $(\beta p_s + (1 - \beta)p_r)R_s \geq 1$) or nobody, i.e., there is financial collapse.